NOTE

Read more about how Google DeepMind partners with Google Maps to predict traffic patterns with Graph Neural Networks here https://deepmind.google/blog/traffic-prediction-with-advanced-graph-neural-networks/.

REAL WORLD EXAMPLE: A DIGITAL TWIN OF YOU

It is not just machinery and infrastructure that can be modeled in the digital world, but ourselves too. Every time you browse a streaming service or shop online, you're providing data about yourself to systems that are essentially personal predictive digital twins. These systems build and continuously update a profile of your preferences, beliefs, habits, and behaviors. Based on this profile, they predict which item to recommend on Amazon, what music to queue on Spotify, what search results to return to you in Google, or what you might want to watch next on Netflix as shown in figure 1.9.

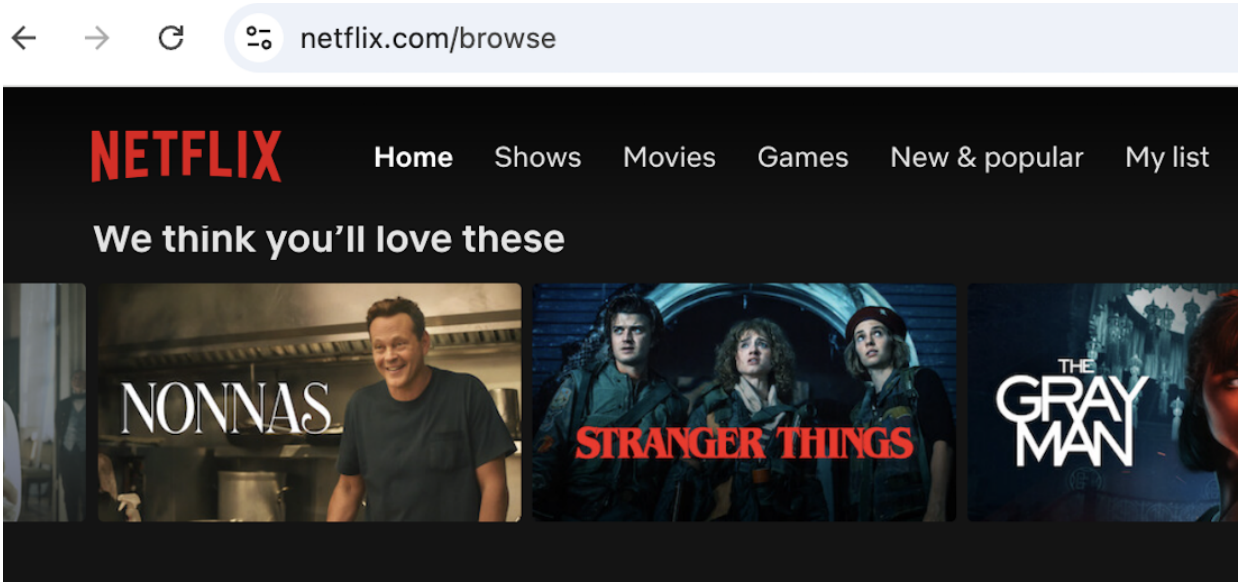


Figure 1.9 What Netflix thinks I would like to watch next based on its representation of my preferences learned through past shows I have watched.

The proliferation of these human-centric twins makes robust privacy protocols and clear ethical guidelines essential to prevent misuse,